

Amazon Customer Support AI Agent

Hiver SDE Intern — Take-Home Assignment Report

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Brand: AmazonHelp

Approach: Intent classification + intent-aware historical retrieval + escalation policy + deterministic response drafting

Executive Summary

This project turns noisy real-world Amazon customer-support conversations into a runnable support-agent pipeline. The system classifies an incoming customer message into one of 12 intents, retrieves historically similar Amazon support cases, decides whether to auto-handle or escalate, and drafts a response when automated handling is considered appropriate.

The implementation intentionally avoids a paid LLM API. A Sentence-Transformer embedding model is used for semantic representations and a balanced Logistic Regression classifier is used for intent prediction. Historical responses are used as retrieval evidence, while the final response layer is deterministic and intent-specific. This keeps the system reproducible and predictable, but also limits response specificity.

The strongest measured headline result is **55.83% intent accuracy on a 240-message manually labelled golden set**. A trivial majority baseline achieved 29.17%, while a TF-IDF + Logistic Regression baseline achieved 52.08%. The semantic classifier therefore improves over the trivial baseline by 26.67 percentage points and over the simple TF-IDF baseline by 3.75 points.

The conclusion is deliberately conservative: this is a working proof-of-concept, not a production-ready autonomous support agent. The evaluation exposes ambiguity between delivery, tracking, support-complaint, and product/app issues, so the report emphasizes failure analysis and next steps.

Hiver asks for a runnable repository, a 150–250 example hand-labelled golden set, automated evaluation plus an LLM-as-judge rubric with human agreement, a report of at most six pages, and a decision log.

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1. Problem Framing

What does “good” mean for Amazon customer support?

- Correctly recognize the customer's main support issue without over-interpreting ambiguous wording.
- Use historical Amazon support behavior as evidence rather than inventing unsupported resolutions.
- Avoid unsafe automation for account/security problems, explicit human-support requests, and complex complaints.
- Produce concise, actionable responses for routine issues.
- Be transparent about uncertainty and prefer escalation when the system cannot confidently support automation.

What was not built?

- No production chat UI, authentication system, ticketing integration, or live Amazon API integration.
- No paid LLM API dependency and no fine-tuning of a large language model.
- No attempt to run the full ~3M-row source dataset at runtime; a bounded historical subset is used.

The assignment permits an LLM API or open model and explicitly expects/encourages a dataset subsample rather than running the full source dataset. ■filecite■turn57file2■L117-L123■ ■filecite■turn57file2■L145-L150■

2. Data and Intent Design

AmazonHelp was selected as the single brand. The data was cleaned by removing missing/empty records, duplicate customer-response pairs, extremely short messages, acknowledgement-only messages, and non-English conversations. A final historical dataset and a manually labelled golden set were kept separate from classifier training and retrieval where applicable.

The final taxonomy contains 12 intents:

Intent	Definition
Order Tracking / Status	Where is the order, tracking or current shipment status.
Delivery Delay	Late, missed, failed or unexpectedly delayed delivery.
Delivered but Not Received	Tracking says delivered but customer did not receive it.
Wrong / Damaged / Defective Item	Wrong, damaged, defective or counterfeit product.
Return / Refund	Return request, refund request or refund status.
Order Cancellation	Request to cancel an order.
Payment / Billing	Charges, billing problems or unexpected payment issues.
Prime Membership / Benefits	Membership status, eligibility or benefits; charges are treated as billing.
Account / Login / Security	Login, account access or security concerns.
Product / App / Website Issue	Technical or functional issue with Amazon product/app/site.
Customer Support / Complaint	The support interaction itself is the primary complaint.
Non-actionable / Exclude	Greeting, thanks, vague statement or no actionable support request.

3. System Design

Pipeline: customer message → intent classification → semantic retrieval → intent-aware reranking → escalation decision → response drafting.

Intent classifier. The final classifier uses Sentence-Transformer *all-MiniLM-L6-v2* embeddings followed by balanced Logistic Regression. It was trained on 1,317 labelled examples across all 12 intents. The manually labelled 240-message golden set was kept out of training.

Historical retrieval. Up to 10,000 cleaned Amazon historical cases are embedded. The system retrieves the top 50 semantic candidates and softly reranks them using predicted intent agreement. The final score is $0.80 \times \text{semantic similarity} + 0.20 \times \text{intent match}$.

Escalation policy. The system escalates low-confidence predictions, weak retrieval, account/security issues, explicit human-support requests, Customer Support/Complaint, and Non-actionable messages. This conservative policy intentionally trades automation rate for safety.

Response layer. Responses are deterministic templates by intent. They provide predictable actions and avoid unsupported policy details. The trade-off is lower specificity than free-form LLM generation.

4. Results

Evaluation	Result
Golden intent accuracy	55.83% (134 / 240)
Trivial majority baseline	29.17% (70 / 240)
TF-IDF + Logistic Regression	52.08% (125 / 240)
Semantic + Logistic Regression	55.83% (134 / 240)
Semantic gain over trivial	+26.67 percentage points
Semantic gain over TF-IDF	+3.75 percentage points
Retrieval top-1 intent consistency	55.42% (133 / 240)
Human response review	50 responses; overall mean 3.64 / 5
Human response relevance	4.52 / 5
Human response actionability	2.12 / 5
Human response safety	5.00 / 5
Human response grounding	3.86 / 5
Human responses rated 4–5 overall	76% (38 / 50)

The human response review is a manual review of 50 sampled responses. It is useful evidence about response quality, but it is **not** an LLM-as-judge evaluation.

5. Baselines

The trivial baseline always predicts the majority golden-set intent (Delivery Delay). The simple baseline is TF-IDF with Logistic Regression. The final semantic model uses sentence embeddings with Logistic Regression. The comparison shows a modest improvement over the lexical model and a large improvement over the trivial baseline.

6. Top Five Failure Modes

1. Intent classification confusion — 106/240 golden examples were misclassified. The largest single confusion was Delivery Delay → Delivered but Not Received (11 cases). Hypothesis: delivery language is highly overlapping, and short messages often omit the evidence needed to distinguish “late” from “marked delivered.”

Real example: “my order is still not fucking delivered, I want a fucking proper service” was classified as Delivered but Not Received even though there is no explicit delivered status.

2. Wrong-intent retrieval — 107/240 top-1 retrievals did not match the golden intent. Semantic similarity can rank related delivery, tracking, refund or support cases above the best issue-specific example.

Real example: a delayed-delivery message retrieved a case with very similar delivery wording but a different underlying issue before intent-aware reranking.

3. Over-escalation of routine cases — 112 expected AUTO_HANDLE cases were escalated. The dominant trigger was low classification confidence. This makes the system safer but reduces automation usefulness.

Real examples: routine delivery-delay messages were escalated solely because classifier confidence was below 0.50.

4. Missed escalation on complex/risky cases — 8 expected ESCALATE cases were auto-handled. Confidence alone does not reliably detect fraud-like or investigation-heavy situations.

Real example: “My last package went missing while you people marked final status as ‘Delivered’ It was a clear fraud and has happened with many known ones.” was auto-handled despite the fraud/investigation signal.

5. Response grounding and specificity limitation — the response layer uses deterministic intent templates rather than generating each reply directly from retrieved historical evidence. This keeps outputs safe and reproducible, but responses can be generic. The manual review’s lower actionability score (2.12/5) is consistent with this limitation.

7. What Is Misleading About My Headline Number?

The headline classification accuracy is **55.83%** on the 240-message golden set. It would be misleading to present this as “the agent resolves 55.83% of customer issues.” Classification is only one stage; retrieval, escalation, and response quality introduce additional failure points.

The golden set is also intentionally small and manually labelled. Some intent boundaries are subjective, especially Delivery Delay vs Order Tracking and Delivery Delay vs Delivered but Not Received. The number is evidence of useful signal, not a production success rate.

The current response evaluation is human-reviewed rather than the assignment’s requested LLM-as-judge-with-human-agreement setup. I do not claim that requirement is fully satisfied in this version. This is an explicit limitation rather than a hidden metric.

8. What I Would Do With One More Week

- Increase the manually labelled training set, focusing on the highest-confusion intent pairs.
- Improve historical intent annotation and retrieval reranking.
- Tune escalation thresholds using a dedicated policy-labelled validation set.
- Add a risk/complexity classifier for fraud, account takeover, repeated failed support, and investigation-heavy cases.
- Generate responses from retrieved historical evidence using a reliable open model or approved LLM API, with evidence constraints.
- Build a proper LLM-as-judge rubric and measure judge/human agreement.
- Add a lightweight Streamlit demo only after the evaluation pipeline is stable.

9. Decision Log

Decision	Why
Choose AmazonHelp	Large practical single-brand support pool for a focused prototype.
Use one brand only	Keeps intents and historical resolution behavior coherent.
Create 12 intents	Covers major support patterns without creating an overly fragmented taxonomy.
Keep Non-actionable / Exclude	Prevents greetings and vague messages from being forced into support issues.
Treat Prime charges as Billing	The root issue is a financial charge, not membership benefits.
Keep golden set separate	Reduces evaluation leakage and protects the headline metric.
Use semantic embeddings	Support language is noisy and paraphrastic; lexical matching alone is insufficient.
Use Logistic Regression	Fast, simple, interpretable and reproducible on top of embeddings.
Use soft intent-aware retrieval	Hard filtering would amplify classifier errors.
Bound history at 10,000 rows	Keeps runtime practical while retaining substantial historical evidence.
Use conservative escalation	Uncertainty and sensitive account/security cases should not be blindly automated.
Avoid paid API dependency	Keeps the project runnable without API credits or quota failures.
Use deterministic responses	Predictable and safer than unsupported free-form generation in the no-API version.
Report failures openly	The assignment prioritizes proof, not only best-looking metrics.

10. Reproducibility and Submission Notes

The repository is structured around a runnable entry point (**python run_agent.py**), source scripts, processed data, evaluation outputs, and this report. The README contains installation and execution instructions. The full original Kaggle dataset is not included because of its size; the processed files needed for the prototype and evaluation are included.

The assignment requires a runnable repository, a 150–250 example golden set, an evaluation harness, a report and a decision log. The current project provides the 240-message manually labelled intent golden set, automated evaluation outputs, manual response review, report and decision log. The remaining evaluation limitation is the absence of a reliable LLM-as-judge plus measured judge/human agreement.

11. Final Takeaway

This prototype demonstrates the complete engineering loop: noisy data preparation, brand-specific intent discovery, supervised semantic classification, historical retrieval, conservative escalation, response drafting, baselines, evaluation and failure analysis. The most important result is not the 55.83% number by itself; it is the evidence showing where the system succeeds, where it fails, and what engineering work would most improve trustworthiness next.